DINGA JAVU

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Professional Summary

Biomedical Engineering graduate with an analytical and problem-solving mindset. Skilled in machine learning model development, including deep learning, NLP, regression, ensemble and association learning models. Proven proficiency in building end-to-end data pipelines. Experienced using Amazon Web Services and tools for data preprocessing, model building, deployment, inference and performance monitoring. Detail-oriented approach to Generative AI solution design adhering to responsible data usage. Skilled at leveraging LLMs using LangChain to build solutions such as agents and chatbot assistants tailored to specific business applications.

Work Experience

Storky Consulting and Projects | Trainee Engineer **Sep 2024 – Dec 2025**

Reduced production delays by identifying an opportunity to integrate existing infrastructure with new equipment

Assisted the project manager in providing solutions to optimise interdepartmental communication to reduce bottlenecks in logistical workflows

Learnt how to maintain technical documentation that upheld rigorous standards to provide traceability and accuracy on project progress records

Showed people management skills in managing and solving workforce and management disputes, staving off delays

In a fast-paced, highly pressured environment, I maintained result delivery.

Education

University of Witwatersrand, BEngSc in Biomedical Engineering **2025**

Relevant Coursework:

- Biomedical Statistics & Numerical Methods
- Engineering Mathematics
- Software Development
- Biomedical Measurement, Instrumentation & Imaging

Academic Projects

Designed and built a biomedical diagnosis device and circuit that extracts raw accelerometer signals from finger tapping and processes the signal data to visualise a clean, noise-free amplitude signal intended to detect tremors

Designed and implemented a 3-element Windkessel model in MATLAB to simulate haemodynamic behaviour in the aortic system, using signal and control system principles to observe the effects of pathological conditions on the systolic and diastolic blood pressure

Designed a solution in MATLAB leveraging machine learning principles in image classification to detect pathologies in erythrocyte slides

Languages and Computer Tools

Languages:

- English (Fluent)
- IsiXhosa (Fluent)
- Afrikaans (Fluent)
- IsiZulu (Fluent)
- Sesotho (Intermediate)

Computer Tools:

- Python
- R
- C++
- AWS Cloud Services
- LangChain
- GitHub
- Microsoft Excel
- MySQL
- PostgreSQL
- LTSpice
- MATLAB

Independent Projects - Online Portfolio; click to access

CNN Breast Cancer Image Classifier

Designed and trained a custom convolutional neural network on 277,524 histopathological image patches

Evaluated metrics in a clinically aligned context

Built a rigorous data pipeline that ensured patient-level slide splitting and class-imbalance handling

Grad-CAM visualisations to evaluate which microscopic tissue regions the model used to make diagnosis decisions

Clinical Guidelines Q&A Assistant

Built a retrieval-augmented generation pipeline over clinical literature and guidelines, combining vector search with LLM-based (Google Gemini API) synthesis to yield educated, source-cited answers

IDC Grad-CAM Explainer AI Tool

Built a clinical AI-powered tool designed to support an oncological professional clinician, not diagnose. Upon inference on an input histopathological patch of breast tissue, the solution classifies the uploaded patch as benign or malignant (predicting the presence of Invasive Ductal Carcinoma).

A Grad-CAM heatmap overlay is then displayed to indicate which regions of the tissue contributed most to the model's classification to aid the clinician in their analysis.

An LLM's power is then leveraged to explain the output as well as giving the user (clinician) an opportunity to ask follow-up questions in a conversational format with history. The LLM is tailored to provide honest information available to it, leaving no room for hallucinated answers based on confidently fabricated information.

Skills & abilities

- Problem-Solving
- Structured Analytical Thinking
- Stakeholder Engagement
- Prompt Engineering
- Feature Engineering
- Experimental Design
- Model Versioning
- Query Optimisation
- Vector Databases
- Documentation
- Data Pipeline Architecture
- Circuit Design and Analysis

References

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